Project title: EduGenie -Google Gemini Powered Learning Assistant

Phase 2: Requirement Analysis Phase

1. Requirement Analysis

The requirement analysis phase focuses on identifying all the requirements needed to develop and operate the EduGenie application. Both functional and non-functional requirements were considered during this phase.

Functional Requirements:

-The system should allow students to interact with the learning assistant and enter their questions or learning-related requests. The application should process the user's input and generate an appropriate response using the Gemini AI model.

The major functional requirements include:

-The user should be able to enter a question or learning request.

-The system should process the entered text.

-The application should communicate with the Google Gemini AI model.

-The system should generate an AI-based response.

-The response should be displayed to the user.

-The system should support learning-related explanations.

-The application should support quiz-related activities.

-The system should provide summarized learning content where required.

-The system should handle errors without crashing.

Non-Functional Requirements

The application should be easy to use and should provide responses within a reasonable amount of time. The system should also maintain proper security for sensitive configuration information such as the Gemini API key.

Important non-functional requirement include:

-User-friendly interface
-Good response time
-Reliability
-Security
-Maintainability
-Scalability
-Error handling
-Easy deployment and execution

Technologies Required:

The major technologies used for the project include Python, FastAPI, Google Gemini AI, HTML, CSS, JavaScript, and Visual Studio Code.